

Chapter 4: Web Technology

4.1 Concepts of Web Technology

Introduction

In today's digital world, the internet has become an important part of our daily life. We use websites for learning, communication, entertainment, shopping, banking, and many other purposes. Websites are created using different tools, programming languages, and technologies collectively known as **Web Technology**.

What is Web Technology?

Web Technology refers to the collection of tools, software, programming languages, protocols, and techniques used to create, develop, and manage websites and web applications on the internet.

It helps developers create attractive, interactive, and user-friendly websites.

Features of Web Technology

- Helps in creating websites and web applications
- Supports communication and information sharing
- Makes websites interactive and responsive
- Allows multimedia integration such as images, audio, and videos
- Enables online services like e-commerce, online learning, and social media

Components of Web Technology

Web technology mainly consists of:

Component	Description
HTML	Used to create the structure of webpages
CSS	Used to style and design webpages
JavaScript	Used to add interactivity
Database	Stores website data
Web Server	Delivers webpages to users

Examples of Web Technologies

Technology	Purpose
HTML	Structures webpages
CSS	Styles webpages
JavaScript	Adds interactivity

PHP	Server-side scripting
Python	Backend web development
MySQL	Database management
WordPress	Website management system

Advantages of Web Technology

- Easy access to information
- Global communication
- Online learning opportunities
- Business promotion and e-commerce
- Entertainment and social networking

Future of Web Technology

The future of web technology includes:

- Artificial Intelligence (AI)
- Progressive Web Apps (PWA)
- Virtual Reality (VR) and Augmented Reality (AR)
- Better web security
- Faster and more responsive websites

4.2 Concept of UI/UX

Introduction

UI and UX are important parts of modern web design. They help developers create websites and applications that are attractive, easy to use, and user-friendly.

Front End and Back End

Front End

The **Front End** is the visible part of a website or application that users interact with directly.

Examples

- Buttons
- Menus
- Images
- Text
- Layouts

Front End Technologies

- HTML
- CSS
- JavaScript

Back End

The **Back End** is the hidden part of a website that handles:

- Data processing
- Server operations
- Database management
- Business logic

Backend Technologies

- PHP
- Python
- Node.js
- MySQL

UI (User Interface)

Definition

UI (User Interface) refers to the visual elements of a website or application that users interact with.

UI Includes

- Buttons
- Icons
- Colors
- Layouts
- Menus

Objectives of UI

- Make the website attractive
- Improve usability
- Provide easy navigation

UX (User Experience)

Definition

UX (User Experience) refers to the overall experience a user gets while using a website or application.

It focuses on:

- User satisfaction
- Ease of use
- Accessibility
- Smooth interaction

Goals of UX

- Improve user satisfaction
- Make websites easy to use
- Reduce confusion
- Increase engagement

Difference Between UI and UX

UI	UX
Focuses on appearance	Focuses on user experience
Deals with colors, buttons, layout	Deals with usability and satisfaction
Makes website attractive	Makes website easy to use

Advantages of UI/UX Design

- Improves user satisfaction
- Makes websites attractive
- Increases user engagement
- Saves development time and cost
- Helps in branding

Software Used for UI/UX Design

Software	Purpose
Figma	Collaborative UI/UX design
Adobe XD	Interface design
Balsamiq	Wireframe design
Sketch	UI design

Concept of Wireframe

Definition

A **Wireframe** is a simple outline or blueprint of a webpage or application interface.

It shows:

- Placement of elements
- Structure of the page
- Navigation flow

without adding colors or detailed designs.

Importance of Wireframe

- Helps in planning webpage layout
- Improves communication between designers and developers
- Saves development cost
- Identifies design problems early

Wireframe Design

Definition

Wireframe Design is the process of creating a basic visual structure of a webpage or application.

Key Features of Wireframe Design

1. Structural Framework
2. User Flow and Navigation
3. Quick Modification
4. Prototyping
5. Cost-effective Design

Benefits of Wireframe

- Easy to draw
- Easy to understand
- Easy to modify
- No coding required

4.3 HTML

4.3.1 Introduction to HTML

Definition

HTML (HyperText Markup Language) is the standard markup language used to create webpages.

- **HyperText** means text containing links.
- **Markup Language** means a language that uses tags to structure content.

HTML provides the basic structure or skeleton of a webpage.

Developed By

HTML was developed by Tim Berners-Lee in 1991.

Features of HTML

- Easy to learn
- Platform independent
- Supports multimedia
- Supports hyperlinks
- Used to create webpages

Uses of HTML

- Creating webpage structure
- Formatting text
- Adding images and videos
- Creating tables and forms
- Linking webpages

HTML Editors

Editor	Description
Notepad	Basic text editor
Sublime Text	Advanced code editor
Visual Studio Code	Popular web development editor
Notepad++	Lightweight code editor

4.3.2 HTML Tags

Definition

HTML tags are predefined keywords enclosed within angle brackets < > used to define webpage elements.

Example

```
<p>This is a paragraph.</p>
```

Types of HTML Tags

1. Paired (Container) Tag

These tags have both opening and closing tags.

Example

```
<b>Bold Text</b>
```

2. Unpaired (Empty) Tag

These tags do not require closing tags.

Example

```
<br>  
<hr>  
<img>
```

HTML Attributes

Definition

Attributes provide additional information about HTML elements.

Example

```
<font color="red">Hello</font>
```

Common Attributes

- color
- size
- width
- height

- align
- href
- src

4.3.3 Structure of HTML

Basic Structure of HTML Document

```
<!DOCTYPE html>
<html>
<head>
  <title>My Webpage</title>
</head>

<body>
  <h1>Welcome to HTML</h1>
  <p>This is a webpage.</p>
</body>

</html>
```

Main Parts of HTML Structure

Tag	Purpose
<html>	Root element
<head>	Contains webpage information
<title>	Displays page title
<body>	Contains visible webpage content

Comment Tag

Syntax

```
<!-- This is a comment -->
```

Uses

- Adds notes in code
- Helps developers understand code
- Ignored by browser

Paragraph Tag

```
<p>This is a paragraph.</p>
```

Line Break Tag

`<br>`

Used to move text to a new line.

Horizontal Line Tag

`<hr>`

Used to draw a horizontal line.

4.3.4 Text Formatting Tags

Text formatting tags are used to change the appearance of text.

Tag	Purpose
<code></code>	Bold text
<code><i></code>	Italic text
<code><u></code>	Underline text
<code><sup></code>	Superscript
<code><sub></code>	Subscript

Heading Tags

HTML provides six heading tags.

`<h1>Heading 1</h1>`

`<h2>Heading 2</h2>`

- `<h1>` is the largest
- `<h6>` is the smallest

Example of Text Formatting

`<b>Bold</b>`

`<i>Italic</i>`

`<u>Underline</u>`

H`<sub>2</sub>`O

A`<sup>2</sup>`

Font Tag

Used to change:

- Font size
- Font color
- Font face

Example

```
<font color="red" size="5">  
Hello  
</font>
```

Marquee Tag

Used to create scrolling text.

Example

```
<marquee>Welcome</marquee>
```

4.3.5 Anchor, List, Table, Image Tag and Their Properties

Anchor Tag

Definition

The anchor tag is used to create hyperlinks.

Syntax

```
<a href="https://example.com">  
Visit Site  
</a>
```

Attributes of Anchor Tag

Attribute	Purpose
Href	Specifies webpage URL
Target	Specifies where to open link

Target Values

- _blank
- _self
- _parent
- _top

Types of Links

External Link

Links to another webpage.

Internal Link

Links within the same webpage.

HTML Lists

Types of Lists

List Type	Tag
Ordered List	
Unordered List	
Description List	<dl>

Ordered List Example

```
<ol>  
<li>Apple</li>  
<li>Mango</li>  
</ol>
```

Unordered List Example

```
<ul>  
<li>Apple</li>  
<li>Mango</li>  
</ul>
```

Description List Example

```
<dl>  
<dt>HTML</dt>  
<dd>Markup Language</dd>  
</dl>
```

Image Tag

Definition

The tag is used to insert images.

Syntax

```

```

Attributes of Image Tag

Attribute	Purpose
src	Image path
width	Width of image
height	Height of image
border	Border size
alt	Alternative text

Table Tag

Definition

HTML tables organize data into rows and columns.

Table Tags

Tag	Purpose
<table>	Creates table
<tr>	Table row
<th>	Table heading
<td>	Table data

Example of Table

```
<table border="1">
```

```
<tr>  
<th>Name</th>  
<th>Age</th>  
</tr>
```

```
<tr>  
<td>Arom</td>  
<td>20</td>  
</tr>
```

```
</table>
```

Table Attributes

Attribute	Purpose
border	Border thickness
width	Table width
height	Table height
bgcolor	Background color
cellpadding	Space inside cells
cellspacing	Space between cells

4.3.6 Form and Div Tag

HTML Form Tag

Definition

The <form> tag is used to collect user information.

Examples of Forms

- Login form
- Registration form
- Contact form

Form Syntax

<form>

</form>

Form Attributes

Attribute	Purpose
name	Form name
method	GET or POST
action	Server processing location

Input Types

Type	Purpose
text	Text input
password	Password input
radio	Single selection
checkbox	Multiple selection

submit	Submit form
reset	Reset form

Text Field Example

```
<input type="text">
```

Password Field Example

```
<input type="password">
```

Radio Button Example

```
<input type="radio" name="gender"> Male
```

Checkbox Example

```
<input type="checkbox"> Football
```

Textarea Example

```
<textarea rows="5" cols="20">  
</textarea>
```

Select Box Example

```
<select>  
<option>Kathmandu</option>  
<option>Pokhara</option>  
</select>
```

Submit and Reset Buttons

```
<input type="submit">
```

```
<input type="reset">
```

DIV Tag

Definition

The <div> tag is a container used to group HTML elements together.

Advantages of DIV Tag

- Organizes webpage layout
- Helps in styling with CSS
- Supports responsive design
- Easy content grouping

Example

```
<div>  
<h1>Hello</h1>  
<p>Welcome</p>  
</div>
```

4.4 CSS

4.4.1 Introduction to CSS

Definition

CSS (Cascading Style Sheets) is a stylesheet language used to control the appearance and layout of webpages.

CSS works together with HTML.

HTML creates structure while CSS adds design.

Features of CSS

- Changes colors and fonts
- Controls webpage layout
- Makes webpages responsive
- Reduces repeated code
- Improves webpage appearance

Advantages of CSS

- Faster webpage styling
- Easy maintenance
- Reusable styles
- Better user experience
- Responsive design support

CSS Syntax

```
selector {  
  property: value;  
}
```

Example

```
p {  
  color: red;  
}
```

Components of CSS

Component	Description
Selector	Selects HTML element
Property	Defines style type
Value	Specifies style value

CSS Selectors

1. Element Selector

```
p {  
  color: blue;  
}
```

2. ID Selector

```
#header {  
  color: red;  
}
```

3. Class Selector

```
.note {  
  color: green;  
}
```

4. Universal Selector

```
* {  
  margin: 0;  
}
```

CSS Box Model

The CSS Box Model consists of:

Part	Description
Content	Actual content
Padding	Space around content
Border	Boundary around element
Margin	Space outside border

4.4.2 Embed CSS Script in HTML

CSS can be added to HTML using:

- Inline CSS
- Internal CSS
- External CSS

4.4.3 Types of CSS

1. Inline CSS

Definition

CSS written directly inside HTML tags.

Example

```
<p style="color:red;">  
Hello  
</p>
```

Advantages

- Easy for small changes
- Quick styling

Disadvantages

- Difficult to manage large websites

2. Internal CSS

Definition

CSS written inside <style> tag within <head> section.

Example

```
<head>

<style>

p {
  color: blue;
}

</style>

</head>
```

Advantages

- Styles multiple elements
- Easy for single webpage styling

3. External CSS

Definition

CSS written in a separate .css file.

HTML File

```
<link rel="stylesheet" href="style.css">
```

CSS File

```
p {
  color: green;
}
```

Advantages

- Reusable styles
- Easy maintenance
- Consistent website design

Difference Between Types of CSS

Inline CSS	Internal CSS	External CSS
Inside HTML tag	Inside <style> tag	Separate CSS file

Used for single element	Used for single webpage	Used for multiple webpages
Difficult to maintain	Moderate maintenance	Easy maintenance

Common CSS Properties

Property	Purpose
color	Text color
background-color	Background color
font-size	Text size
font-family	Font type
text-align	Text alignment
Border	Border styling
Margin	Outer spacing
Padding	Inner spacing

Example of CSS

```
<!DOCTYPE html>
<html>
<head>
<style>
h1 {
  color: blue;
  background-color: yellow;
}
p {
  color: red;
}
</style>
</head>
<body>
<h1>Hello World</h1>
<p>This is CSS example.</p>
</body>
</html>
```